Keywords: zeroth-order optimization • memory-efficient fine-tuning • variational multiscale methods • LLM adaptation

ENGINE: Variational Multi-Scale Zeroth-Order LLM Fine-Tuning

A Closed-Form Subgrid Correction from Computational Mechanics Yields 2.61× Faster ZO Convergence at Zero Memory Overhead

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arXiv:2611.03293 • cs.LG • Submitted 2 Nov 2026

ENGINE applies the Variational MultiScale (VMS) decomposition from computational fluid mechanics to zeroth-order LLM fine-tuning. By decomposing perturbations into a low-rank coarse component and computing a closed-form subgrid correction that reintroduces unresolved curvature at no additional cost, ENGINE achieves a 2.61× speedup in forward passes over prior ZO methods with zero additional peak memory, and consistently outperforms state-of-the-art ZO baselines on NLP benchmarks.

AT A GLANCE

Problem: Zeroth-order optimization for LLM fine-tuning is slow to converge because low-rank perturbation subspaces miss curvature from the complement space.

Why it matters: ZO fine-tuning is the only way to train LLMs with RL rewards or in memory-limited settings without backpropagation; improving its efficiency makes it practical.

Method: VMS decomposes the parameter space into a low-rank coarse component (updated with 2 forward passes) and a fine-scale complement; a closed-form subgrid correction reintroduces complement curvature with no extra forward passes.

Result: 2.61× speedup in forward passes over LOREN; zero additional peak memory overhead; consistent improvements on SuperGLUE, MMLU, and instruction-following.

Evidence: GPT-J-6B on SuperGLUE, Llama-3-8B on MMLU, and instruction-following on Alpaca Eval.

The Big Picture

Zeroth-order (ZO) optimization enables fine-tuning LLMs without backpropagation, reducing memory from $O(\text{parameters})$ to $O(1)$ for the optimizer state. This matters in three settings: (1) fine-tuning on GPUs with insufficient VRAM for gradients, (2) fine-tuning through black-box API access where gradients are unavailable, and (3) fine-tuning with non-differentiable reward signals such as human feedback or unit-test pass rates.

The main limitation is convergence speed: estimating gradients by measuring reward differences between perturbed parameters is noisy. Restricting perturbations to a low-rank subspace (as in MeZO and LOREN) reduces noise but introduces a systematic bias: curvature from the complement of the subspace is ignored, slowing convergence.

ENGINE addresses this with an insight from computational mechanics: the unresolved fine-scale curvature can be reintroduced via a closed-form correction derived from the coarse (low-rank) gradient estimate, without any additional forward passes.

Why Existing Approaches Fall Short

Full-space ZO (MeZO) estimates gradients with random directions in the full parameter space. The variance is $O(d)$ where d is the parameter count, making it impractically noisy for billion-scale models.

Low-rank ZO (LOREN) restricts perturbations to a rank-r subspace. This reduces variance to $O(r)$ but discards curvature from the $d-r$ dimensional complement, introducing a systematic bias that slows convergence. With $r \ll d$, the complement curvature is large.

Increasing the rank r reduces bias but increases memory and noise proportionally. There is no free way to access the complement curvature within the two-forward-pass budget.

Core Mechanism: Variational MultiScale Decomposition

The VMS framework decomposes the parameter space into a coarse subspace $\mathcal{U}$ (spanned by a low-rank basis $U \in \mathbb{R}^{d\times r}$) and its complement $\mathcal{U}^{\perp}$. Any perturbation z decomposes as:

$$z = z_c + z_f, \quad z_c = UU^{\top} z, \quad z_f = (I - UU^{\top})z$$

ENGINE estimates the coarse gradient using the standard two-point ZO estimator restricted to z_c:

$$\hat{g}_c = \frac{\ell(\theta + \mu z_c) - \ell(\theta - \mu z_c)}{2\mu} \cdot z_c$$

using 2 forward passes. The **subgrid correction** δ_f accounts for the fine-scale contribution:

$$\delta_f = (I - UU^{\top}) H_{\text{diag}}\, \hat{g}_c$$

where H_{diag} is a diagonal approximation of the Hessian maintained as an exponential moving average of squared gradient magnitudes. This term has zero additional forward-pass cost and $O(d)$ memory, equal to the existing optimizer state.

The final update is:

$$\theta \leftarrow \theta - \eta\big(\hat{g}_c + \alpha\,\delta_f\big)$$

where α is a correction weighting hyperparameter.

Learning or Inference Procedure

Initialization: Compute U as the top-r right singular vectors of a batch of initial ZO gradient estimates. Maintain H_{diag} as an exponential moving average of $\hat{g}_c^2$.

Per-step update:

1. Sample $z_c \sim \mathcal{N}(0, UU^\top)$ (a low-rank direction).
2. Evaluate $\ell(\theta \pm \mu z_c)$ with 2 forward passes.
3. Compute $\hat{g}_c$ and subgrid correction δ_f.
4. Update θ using $\hat{g}_c + \alpha\,\delta_f$.
5. Update H_{diag} with the new squared gradient components.

Periodic basis refresh: Every $K = 200$ steps, recompute U from recent gradient history via a lightweight SVD of a $d \times K$ matrix of gradient directions.

What the Guarantee Says

Proposition 1 (informal): The expected bias of the ENGINE gradient estimate is strictly lower than that of LOREN with the same rank r, provided $\alpha > 0$ and H_{diag} approximates the diagonal Hessian. No exact convergence rate is derived, but the proposition bounds the bias reduction as a function of the quality of the diagonal Hessian approximation.

The practical speed-up guarantee is empirical, not asymptotic. No formal convergence rate is provided in the paper.

Experimental Findings

- **SuperGLUE (GPT-J-6B):** ENGINE matches or exceeds full-fine-tuning accuracy on 7 of 8 tasks; achieves the same accuracy as LOREN with **2.61× fewer forward passes**.
- **MMLU (Llama-3-8B):** ENGINE achieves 72.3% 5-shot accuracy, vs. 69.1% for LOREN and 70.8% for MeZO at the same forward-pass budget.
- **Alpaca Eval:** ENGINE win-rate vs. LOREN is 61.4%, confirming quality improvements in open-ended instruction following.
- **Memory:** Peak memory overhead of ENGINE vs. MeZO is 0%—the diagonal Hessian state fits within the existing optimizer state memory.

Ablations and Interpretation

Without subgrid correction ($\alpha = 0$): Performance equals LOREN, confirming that the speedup is entirely due to the correction.

Rank r sensitivity: ENGINE with $r = 32$ outperforms LOREN with $r = 128$ on SuperGLUE, demonstrating that the correction compensates for a fourfold reduction in rank.

Hessian approximation quality: Using a richer block-diagonal Hessian estimate improves convergence by 7% but requires 1.3× more memory. The diagonal approximation provides most of the benefit.

Basis refresh interval K: Performance is robust to K in the range 100–500, suggesting that the low-rank basis does not need to be updated frequently.

Reference

Zhuoli Ouyang, Bowen Li, Chen Hao, Yejin Choi. **ENGINE: Endogenous Variational MultiScale Optimization for Zeroth-Order LLM Fine-Tuning.** arXiv:2611.03293. 2026. https://arxiv.org/abs/2611.03293